

Interactions and mechanisms in iron-fortified skim milk: Physicochemical properties, calcium bioavailability, and cellular responses

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ABSTRACT

Iron fortification of skim milk offers a promising strategy to address global micronutrient deficiencies, however, its effects on the physicochemical properties, mineral distribution, and bioavailability of milk remains poorly understood. The present study provides a comprehensive analysis of the interactions and mechanisms involved in iron-fortified skim milk systems. Iron salts (ferrous chloride, iron(II) lactate, and iron(II) gluconate) induced a concentration-dependent pH reduction, with ferrous chloride causing the most significant acidification. Iron distribution varied by salt type, with iron(II) gluconate showing higher serum iron levels. Calcium redistribution revealed a 22–28 % aqueous-phase depletion due to competitive iron-phosphate binding. Particle size remained stable, while ζ -potential decreased, reflecting iron-induced charge neutralization. Viscosity increased with iron concentration, indicating enhanced casein cross-linking. In vitro digestion and cellular uptake studies demonstrated that skim milk protein hydrolysates (MH) improved calcium bioavailability by forming soluble peptide-calcium complexes, upregulating calcium transporters (TRPV6 and Claudin-2) and mitigating iron-induced oxidative stress. MH also enhanced antioxidant activity, reducing ROS and lipid peroxidation while upregulating ferroportin (FPN1) via Nrf2 activation. These findings highlight skim milk as an effective carrier for iron fortification, enabling simultaneous calcium and iron supplementation while maintaining stability and bioavailability.

1. Introduction

Iron and calcium are two essential micronutrients that play critical roles in human health. Iron is a vital component of hemoglobin, myoglobin, and various enzymes, making it indispensable for oxygen transport, energy production, and cellular metabolism (Piskin et al., 2022). Calcium, on the other hand, is crucial for bone health, muscle function, nerve transmission, and blood clotting (Pu et al., 2016). Despite their importance, deficiencies in these minerals remain prevalent worldwide, particularly in vulnerable populations such as children, pregnant women, and the elderly (Cormick & Belizán, 2019; Gernand et al., 2016; Tang & Sholzberg, 2024). Iron deficiency anemia and calcium deficiency disorders, such as osteoporosis, are significant public health concerns, necessitating effective strategies for improving dietary intake and bioavailability of these nutrients (Geissler & Singh, 2011). Milk and dairy products are widely recognized as excellent sources of calcium due to their high mineral content and bioavailability (Fairweather-Tait & Teucher, 2002). Skim milk, in particular, is a

popular choice for fortification due to its low-fat content and widespread consumption. However, the fortification of skim milk with iron presents several challenges. Iron is known to interact with milk components, such as casein and whey proteins, which can affect its solubility, stability, and bioavailability (Kurtz et al., 1973). Additionally, the addition of iron salts to skim milk can alter its physicochemical properties, including pH, particle size, and zeta-potential, which may influence the overall quality and functionality of the fortified product (Gaucheron, 2000; Raouche et al., 2009). Understanding these interactions is crucial for developing effective iron-fortified dairy products that maintain both nutritional and sensory qualities. The bioavailability of calcium in iron-fortified skim milk is another critical consideration. Calcium and iron are known to compete for absorption in the gastrointestinal tract, which can lead to reduced bioavailability of one or both minerals (Olusanya, 2022). This interaction is particularly relevant in the context of iron fortification, as excessive iron levels may impair calcium absorption, potentially exacerbating calcium deficiency (Li et al., 2025). Therefore, elucidating the mechanisms underlying calcium bioavailability in iron-fortified skim

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milk is essential for optimizing the nutritional benefits of fortified dairy products. In addition to physicochemical and nutritional considerations, the cellular responses to iron-fortified skim milk are an important aspect of this research. Iron fortification can induce oxidative stress in cells due to the pro-oxidant nature of iron, which may lead to cellular damage and impaired function (Núñez et al., 2001). Assessing cell viability and antioxidant activity in the presence of iron-fortified skim milk can provide valuable insights into the potential health impacts of fortified dairy products (Smialowska et al., 2017).

Accordingly, the aim of the present study is to comprehensively investigate the interactions and mechanisms in iron-fortified skim milk, with a focus on physicochemical properties, calcium bioavailability, and cellular responses. Specifically, the research explores the effects of adding ferrous salts on the pH, iron and calcium distribution, particle diameter, zeta-potential, and fluid flow behavior of skim milk. Additionally, the mechanisms of calcium bioavailability in iron-fortified skim milk hydrolysates are examined, along with the buffering effects of milk hydrolysates on calcium absorption. Cellular responses, including cell viability and antioxidant activity, are also assessed to evaluate the potential health impacts of iron fortification. These findings are important for developing iron-fortified dairy products that effectively address nutritional needs while maintaining consumer appeal. By elucidating the complex interactions between iron and milk components, this study improves our knowledge of mineral fortification and absorption. Furthermore, the insights gained from this research can inform the design of functional dairy products that address global micronutrient deficiencies while maintaining product quality.

2. Materials and methods

2.1. Materials

Low heat skim milk powder was purchased from Fonterra Co-operative Group, New Zealand. Iron(II) lactate hydrate (purity $\geq 98\%$), iron(II) D-gluconate dehydrate (purity $\geq 98\%$), ferrous chloride tetrahydrate (purity $\geq 98\%$), porcine pepsin (P-7012), porcine pancreatin (P-1750) and bile salts (48305) were all purchased from Sigma-Aldrich (Shanghai, China). All other chemicals and reagents used in this study were of analytical grade and used as received, without further purification.

2.2. Preparation of iron-fortified skim milk and physicochemical characterization

2.2.1. Milk preparation and iron fortification

Low-heat skim milk powder was reconstituted in deionized water at 13 % (w/w) and 25 °C for 2 h, followed by overnight storage at 4 °C to ensure complete protein hydration. Sodium azide was added to the milk at 0.02 % (w/w) as a preservative (Raouche et al., 2009). Skim milk was then individually fortified with ferrous chloride (FeCl_2), iron(II) lactate (FeLac_2), or iron(II) D-gluconate (FeGlu_2) to achieve final iron concentrations of 1, 5, 10, and 15 mmol/L. Subsequently, deionized water was added to each sample to adjust the total solids concentration to 10 % (w/w). These resulting samples are referred to as iron-fortified skim milk.

2.2.2. pH measurement and adjustment for skim milk samples

The pH of the iron-fortified skim milk samples was measured at 25 °C using the pH meter equipped with a pH electrode, 30 min after the addition and stirring of the iron salts. To minimize the influence of iron salt addition on the inherent pH of the skim milk, samples were adjusted to a pH of 6.8 ± 0.1 using 0.5 mol/L sodium hydroxide. The samples were stirred at 25 °C for 1 h during the pH adjustment, and then stored overnight in a chiller at 4 °C to inhibit microbial growth and allow for a stable mineral equilibrium (Deshmukh et al., 2024; Koutina et al., 2014; Mittal et al., 2015).

2.2.3. Mineral partitioning

All samples were separated into micellar and soluble phases by ultracentrifugation at $100,000 \times g$ for 1 h at 20 °C. Following ultracentrifugation, the supernatant (soluble phase) was carefully aspirated (Mittal et al., 2015). The calcium content of the soluble phase was determined by EDTA titration (Vavrusova et al., 2014), and the iron content by KMnO_4 titration (Allen, 1914). The total phosphorus content in the supernatant was quantified using a total phosphorus content assay kit (Solarbio). Calcium ion activity ($\alpha_{\text{Ca}^{2+}}^2$) was measured at 25 °C using a calibrated calcium ion-selective electrode. The electrode was calibrated with standard calcium chloride solutions at concentrations of 1.0×10^{-4} , 1.0×10^{-3} , and 1.0×10^{-2} mol/L. The calcium ion activity, representing the free calcium ion concentration, was then calculated from the linear relationship described by the Nernst equation: $\text{pCa} = -\log \alpha_{\text{Ca}^{2+}}^2$.

2.2.4. Particle size and ζ -potential measurements

Hydrodynamically averaged (Z-average) diameters and ζ -potential values were obtained using dynamic light scattering. All milk samples were diluted 1:100 with the corresponding collected ultrafiltrate. The refractive index values of 1.45 and 1.33 for protein and dispersant phase, respectively, at 25 °C at a backscatter angle of 173°. The particle diameter was used to monitor the breakdown of casein micelles following ion exchange treatment (Anema et al., 2005; Mittal et al., 2015; Pougher et al., 2024).

2.2.5. Flow behavior analysis

Flow behavior measurement of iron-fortified skim milk samples were carried out using a rheometer (Discovery HR-3, TA Instruments, USA) with a 60 mm parallel plate. A gap height of 1000 μm was used, and silicone oil was applied to the sample edge to prevent evaporation and sample loss. Shear stress was measured over the shear rate range 0.001–100 1/s at 20 °C. Viscosity was also monitored with time in iron-fortified skim milk samples at a constant shear rate of 100 1/s at 20 °C for 120 s (Zhao et al., 2023).

2.3. In vitro simulated digestion and absorption of iron-fortified skim milk

2.3.1. Simulated gastrointestinal digestion

The gastrointestinal in vitro digestion experiments were carried out according to Ménard et al. with minor modifications (Ménard et al., 2018). Considering that the skim milk was liquid and its residence time in the mouth was short, the oral phase was omitted. To prepare iron-fortified skim milk hydrolysates under simulated gastric-proximal intestinal conditions, 20 mL samples were subjected to successive hydrolysis with 10 mg pepsin (2500 U/mg) at pH 2 (2 mol/L HCl), 37 °C for 2 h, followed by 25 mg pancreatin (100 U/mg) with 100 mg bile salt at pH 7.0 (6 mol/L NaOH) at 37 °C for 2 h. The digestion was ended by boiling water bath for 5 min, and after cooling to room temperature, the digest was centrifuged at $10,000 \times g$ for 20 min at 4 °C. The resulting supernatant was then centrifuged at room temperature at 4000g for 15 min using 10 kDa pore-size ultrafiltration membrane tubes to mimic the filtering and shielding effect of the small intestinal mucosa on the digestive fluid. The ultrafiltering fraction was filtered through 0.22 μm polyethersulfone membranes and used as the iron fortified skim milk hydrolysates. To maintain cellular osmotic pressure homeostasis, the filtrate was diluted 4-fold and these samples (Fe-MH) were used in calcium transport experiments (2.3.3–2.3.4 sections). The calcium and iron concentrations in Fe-MH were determined by ICP-OES. In addition, the ultrafiltrate without iron-fortified was desalted by a 1 kDa pore-size dialysis membrane followed by lyophilization at -80°C and 0.04 mbar for 2 days obtaining powder samples to determine the effects of milk hydrolysates (MH) on cellular activity and functional properties (2.3.5–2.3.7 sections).

2.3.2. Cell culture

Caco-2 cells were initially seeded in a 25 cm² gas-permeable flask with 5 mL DMEM supplemented with 10 % fetal bovine serum (FBS) at 37 °C in a 5 % CO₂ humidified incubator. Upon reaching 90–100 % confluence, cells were passaged using 0.25 % trypsin-EDTA for 4 min. The digestion was stopped by adding fresh medium, and cells were pipetted repeatedly to obtain a single-cell suspension (Liu et al., 2023).

2.3.3. Intracellular calcium absorption

Caco-2 cells were seeded at a density of $2\text{--}3 \times 10^5$ cells/mL in 12-well Transwell plates, with 0.5 mL cell suspension in the upper chamber and 1.5 mL medium in the lower chamber. The medium was replaced every other day for the first week, then daily from day 8 onwards. Transepithelial electrical resistance (TEER) was measured using a Millicell-ERS meter to assess monolayer integrity, with a TEER value above 500 $\Omega\cdot\text{cm}^2$ indicating the formation of a tight monolayer. For calcium absorption experiments, cells were washed 2–3 times with calcium- and magnesium-free HBSS. The upper chamber was then filled with 0.5 mL of CaCl₂ or Fe-MH samples (liquid), while the lower chamber contained 1.5 mL of HBSS. After 2 h of incubation at 37 °C, the calcium concentration in the lower chamber was determined by inductively coupled plasma optical emission spectroscopy (ICP-OES). The calcium absorption rate was calculated as the percentage of calcium transferred from the upper to the lower chamber (Liu, Li, & Tang, 2024).

2.3.4. Dynamic measurement of cytosolic Ca²⁺ concentration

Caco-2 cells were seeded in 48-well plates at a density of 7000 cells per well and cultured for 7–8 days in DMEM with 10 % FBS, changing the medium every 2 days. For Fluo-3-AM loading, cells were washed with HBSS to remove serum, then incubated with 10 μM Fluo-3-AM in HBSS for 30 min at room temperature in the dark. After washing twice with HBSS, cells were allowed a 10-min recovery period at 37 °C in the dark for de-esterification (Cui et al., 2015). Calcium measurements were performed by adding CaCl₂ or Fe-MH samples (liquid) to the wells. Fluorescence measurements (excitation/emission: 488/526 nm) were taken at 37 °C using an Infinite 200 PRO plate reader (Tecan) for 2 h. Cytosolic Ca²⁺ influx rates were expressed as fluorescence intensity (F) relative to the mean of the first measurements (F₀), calculated as (F/F₀) and named as (Cegarra et al., 2022).

2.3.5. CCK8 assay

CCK8 assay is used to detect the growth and viability of living cells. Different concentrations of MH were added to determine their effects on Caco-2 cell viability, thus providing suitable sample concentrations for subsequent calcium uptake and analysis experiments (Yago et al., 2024). Caco-2 cells were seeded in 96-well plates at a density of 4×10^4 cells/well and incubated for 48 h. Subsequently, the cell medium was aspirated and replaced with 0.01–2 mg/mL MH (solid dissolved in cell medium). The cells were then incubated for an additional 24 h. After incubation, the sample solution was aspirated and CCK-8 solution was dispensed into each well incubating for 1 h, then the absorbance at 450 nm was measured using a microplate reader.

2.3.6. Measurement of intracellular oxidative damage and antioxidant enzyme activity

Caco-2 cells were cultured in 6-well at a density of 2×10^5 cells/well plates for 48 h until the cells were polymerized to 80 %. Subsequently, 50 μM iron salts with and without 0.3 mg/mL MH (solid dissolved in cell medium containing 1 mM CaCl₂) were incubated for an additional 24 h. The content of protein was determined by BCA protein assay kit, and ROS and MDA content as well as SOD and GSH activities were determined according to the instructions of the reagent vendors.

2.3.7. Real-time quantitative polymerase chain reaction (RT-qPCR)

Caco-2 cells were cultured in 6-well at a density of 2×10^5 cells/well plates for 48 h until the cells were polymerized to 80 %. Subsequently,

50 μM iron salts with and without 0.3 mg/mL MH (solid dissolved in cell medium containing 1 mM CaCl₂) were incubated for an additional 24 h. Total RNA was extracted using an RNAPrep Pure Cell/Bacteria Kit (Tiangen, Beijing, China), and its concentration was determined using a NanoDrop-2000c spectrophotometer (Thermo Scientific, Waltham, MA, USA). Reverse transcription was performed using a NovoScript Plus All-in-one 1st Strand cDNA Synthesis SuperMix (gDNA Purge) Kit (Novoprotein, Jiangsu, China). Quantitative RT-PCR was carried out using NovoStart® SYBR qPCR SuperMix Plus kit (Novoprotein, Jiangsu, China) on a CFX96 Touch Real-Time PCR Detection System (Bio-Rad, CA, USA). At least three biological replicates were analyzed for each sample. Relative gene expression was calculated using the 2^{−ΔCt} method, with β-actin as the housekeeping gene. Fold changes were calculated using the 2^{−ΔΔCt} method, compared to untreated controls (Liu, Ma, et al., 2024; Pfaffl, 2001).

2.4. Statistic analysis

IBM SPSS Statistics 25 and Origin 2021 were used for data analysis and graphical representations. Type of salt and salt concentration were main factors for factorial ANOVA with test for interaction. When differences between treatment effects were significant ($p < 0.05$), a multiple comparison of means was performed by Tukey's test.

3. Results and discussion

3.1. Effect of adding iron salts on the pH of skim milk

The pH of skim milk is illustrated in Table 1. The type of salts, salt concentration and the interaction between them showed significant effects on pH of samples. As the interaction was significant, a multiple comparison of means was performed. All types of iron salts exhibited a concentration-dependent decrease with increasing iron salts concentrations from 1 to 15 mmol/L. At the same concentration level (10–15 mmol/L), ferrous chloride induced the most pronounced acidification, followed by iron(II) lactate and iron(II) gluconate. This discrepancy correlated with the relative basicity of the accompanying anions. While chloride ions exhibit extremely weak basicity, allowing full expression of iron's acidic properties, the weakly acidic anions like lactate and gluconate demonstrate stronger basic characteristics. Specifically, gluconate's lower basicity compared to lactate results in less anion hydrolysis, ultimately yielding a more acidic solution at equivalent molar concentrations (Skibsted, 2016).

This phenomenon was consistent across all three iron salts and can be attributed to the inherent acidity of iron salts and cation exchange processes between Fe²⁺ ions and micelle-bound H⁺. Similar to previous observations with zinc salt supplementation (Zhao et al., 2023), iron salt addition may appear to disrupt the ionic equilibrium of milk through analogous mechanisms, where metallic cation introduction prompts hydrogen ion release and subsequent pH reduction. Prior to subsequent analyses, all iron-fortified milk samples were standardized to pH 6.7–6.8 through readjustment.

3.2. Iron distribution in skim milk with and without addition of iron salts

It is noted that temperature change influences the partition of minerals between the colloidal and serum phases, thus, our results are relevant for the study of milk systems under refrigerated storage. In iron-fortified skim milk samples, phase-partitioning analyses revealed differential cation displacement patterns between colloidal and serum phases in order to identified the influence of iron and calcium distribution (Table 1). Distribution of iron ions in iron-fortified milk was depended on the concentration and type of added iron salts. Soluble iron concentrations demonstrated dose-dependent elevation across all three salts, with micellar iron retention rates (80–95 %) aligning with established literature values (Gaucheron, 2000). Notably, the chelation

Table 1

Effect of adding different type of iron salts on the pH of skim milk and the distribution of soluble calcium, iron, phosphorus and calcium ion activity (α_{Ca}^{2+}) in the supernatant of pH-adjusted samples after ultracentrifugation.

Sample	Added Fe salts (mmol/L)	pH (before adjustment)	Calcium (mmol/L)	Iron (mmol/L)	Total phosphorus (mmol/L)	α_{Ca}^{2+}
Skim milk	0	6.75 ± 0.05 ^j	8.31 ± 0.09 ^{fg}	ND	6.88 ± 0.12	(4.88 ± 0.13) × 10 ⁻⁴
	1	6.64 ± 0.02 ^{hi}	7.38 ± 0.18 ^{def}	0.073 ± 0.015	6.33 ± 0.20	(4.70 ± 0.16) × 10 ⁻⁴
FeCl ₂	5	6.30 ± 0.07 ^{efg}	7.25 ± 0.09 ^{de}	0.838 ± 0.005	4.99 ± 0.14	(5.05 ± 0.03) × 10 ⁻⁴
	10	5.92 ± 0.02 ^{ab}	6.12 ± 0.30 ^{ab}	1.769 ± 0.239	3.05 ± 0.02	(5.96 ± 0.33) × 10 ⁻⁴
	15	5.75 ± 0.07 ^a	5.59 ± 0.35 ^a	2.800 ± 0.283	2.30 ± 0.02	(7.83 ± 0.09) × 10 ⁻⁴
FeLac ₂	1	6.63 ± 0.04 ^{hi}	8.8 ± 0.40 ^g	0.104 ± 0.02	6.37 ± 0.31	(4.87 ± 0.03) × 10 ⁻⁴
	5	6.45 ± 0.07 ^{gh}	7.06 ± 0.09 ^{cde}	0.875 ± 0.177	5.46 ± 0.19	(4.85 ± 0.11) × 10 ⁻⁴
	10	6.20 ± 0.06 ^{def}	6.56 ± 0.09 ^{bcd}	1.875 ± 0.177	3.38 ± 0.07	(5.39 ± 0.06) × 10 ⁻⁴
	15	6.01 ± 0.03 ^{bcd}	6.43 ± 0.09 ^{abcd}	3.063 ± 0.088	2.63 ± 0.05	(6.34 ± 0.14) × 10 ⁻⁴
FeGlu ₂	1	6.62 ± 0.04 ^{hi}	7.94 ± 0.09 ^{efg}	0.125 ± 0.015	6.83 ± 0.36	(4.66 ± 0.68) × 10 ⁻⁴
	5	6.39 ± 0.08 ^{fg}	6.75 ± 0.35 ^{bcd}	0.875 ± 0.018	5.30 ± 0.23	(4.69 ± 0.68) × 10 ⁻⁴
	10	6.14 ± 0.08 ^{cde}	6.13 ± 0.18 ^{abc}	1.875 ± 0.018	3.79 ± 0.09	(4.71 ± 0.17) × 10 ⁻⁴
	15	5.97 ± 0.07 ^{bc}	6.19 ± 0.26 ^{abc}	3.125 ± 0.018	2.73 ± 0.02	(5.83 ± 0.02) × 10 ⁻⁴
Type of salt		*	*	*	*	*
Salt concentration		*	*	*	*	*
Interaction		*	*	NS	NS	NS

ND: not detected, NS: no significant effect ($p > 0.05$); *: significant effect ($p < 0.05$).

Values in the same column with different superscript letters were significantly different.

capacity of iron by casein micelle showed no plateau even at concentrations exceeding 10 mmol/L, while Hegenauer et al. reported the saturation threshold of casein micelle binding with 10–20 mmol/L iron, suggesting potential methodological variations in binding assessments (Hegenauer et al., 1979).

For the skim milk with low added iron salt concentrations (< 10 mmol/L), there was no significant difference ($p > 0.05$) observed in the serum iron concentrations for all iron salts. When the added iron salt concentration was 15 mmol/L, however, serum iron levels exhibited salt-dependent variance: iron(II) gluconate > iron(II) lactate > ferrous chloride, inversely mirroring colloidal phase distributions. The difference may be explained by differential metal-ligand association constants (K) governing complex stability. The smaller the K of a metal complex implies a lower affinity of the anion for the Fe²⁺ cation, therefore, more Fe²⁺ are released into the solution. Increased free Fe²⁺ ions results in more available association with the casein micelles and if they do associate, they will likely end up in the colloidal phase. Iron lactate and iron gluconate require two steps to dissociate and order of association constant is iron(II) gluconate > iron(II) lactate, similar to calcium salt dissociation patterns (Skibsted, 2016). Hydroxycarboxylate ligands (lactate/gluconate) may have negative effects on iron binding with micelle through 1:1 iron complex formation, while chloride's minimal coordination capacity permitted maximal Fe²⁺-casein interaction. Among the proteins in milk (whey proteins and caseins), Fe²⁺ binds mostly to caseins, with binding affinity independent of micellar aggregation state. Caseins, especially α_s ($\alpha_{s1} + \alpha_{s2}$)- and β -CN, contain clusters of phosphoserine residues, provide primary iron coordination sites via oxygen ligands (Gaucheron et al., 1996; Shilpashree et al., 2018). Notably, higher additions of iron salts cause flavor and color changes in skim milk, which require further analysis.

3.3. Calcium distribution in skim milk with and without addition of iron salts

The total calcium concentration in 10 % (w/w) reconstituted skim milk revealed 0.028 ± 0.003 mol/L, with 70.0 ± 0.9 % (0.0199 ± 0.0003 mol/L) sequestered in the micellar phase. These values align with literature reports of 1.097 mg/kg total calcium in skim milk

exhibiting 74 % micellar retention (Ahmad et al., 2009; Mittal et al., 2015). The addition of iron salts reduces serum calcium levels. Both factors and the interaction between type of salt and iron salts concentration showed significant effects on soluble calcium concentration (Table 1). As the interaction was significant, a multiple comparison of means was performed. For the three types of iron salts, higher addition levels resulted in lower serum calcium levels. The results showed that adding more than 5 mmol/L ferrous salts induced significant reduction in calcium content. Moreover, the salt-specific phosphorus depletion gradient (chloride > lactate > gluconate) reflects the differential release of free Fe²⁺ ions, governed by association constants and inversely correlating with complex stability. This explains the varying reductions of soluble calcium and phosphate observed with different iron salts. From the percentage of cations present in the serum of iron-supplemented skim milk, the order of association of cations with casein micelles was Fe²⁺ > Ca²⁺ (Table 1). As the binding affinity of iron to casein is higher than that of calcium to casein, the exchange of iron with calcium in the casein micelle has been advocated (Philippe et al., 2005). However, we found that calcium was not released into the serum phase, even upon the addition of 15 mM iron to milk (Table 1). This paradoxical calcium redistribution despite iron's higher casein affinity (Fe²⁺ > Ca²⁺) necessitates multiscale analysis. To explain the phenomenon of decreasing soluble calcium, several parameters must be considered, including the physicochemical properties of cations, the thermodynamic constants of stability, the solubility of the salts formed and the saturation state of the aqueous phase. These parameters allow for the explanation of metal distribution through mechanisms such as cation-anion coordination thermodynamics, salt solubility equilibria, and aqueous phase saturation dynamics.

Ionic calcium dynamics in the soluble phase were systematically monitored using calcium ion-selective electrode (Ca-ISE) to elucidate iron-induced calcium speciation changes and the results are shown in Table 1. Iron fortification (1–15 mmol/L) triggered dose-dependent calcium ion activity (α_{Ca}^{2+}) elevation in skim milk and in 15 mmol/L showed significant difference. The salt-specific response discrepancy was FeCl₂ > FeLac₂ > FeGlu₂, consistent with the analysis above. This phenomenon reflects metal dynamic equilibrium and competitive ligand redistribution in the soluble phase, where calcium exists as: free Ca²⁺

ions, low-molecular-weight complexes (primarily CaHPO_4 and CaCit^-) and micellar calcium (Holt et al., 1981). The hypothetical mechanisms of ions distribution in iron-fortified skim milk samples can be described in Fig. S1. Physiologically, due to calcium ion activity is the more relevant parameter for absorption, the observed increase in calcium ion activity in iron-fortified milk therefore suggests enhanced potential calcium bioavailability.

The observed reduction in soluble calcium and increase in calcium ion activity can be explained by the disruption of existing equilibria within the milk. In natural skim milk systems, soluble calcium primarily complexes with inorganic phosphate (CaHPO_4) and citrate (CaCit^-), maintaining saturated solutions (Gaucheron, 2005). Iron introduction disrupts this equilibrium through competitive salt formation - particularly iron phosphate precipitation [$\text{Fe}_3(\text{PO}_4)_2$] or binding with micellar CPP. Our results indicate that increased iron addition leads to a concentration-dependent decrease in soluble phosphorus concentration, exhibiting a negative correlation, which supports the formation of inorganic or organic complexes (Gaucheron et al., 1997; Kurtz et al., 1973). Among these, ferrous chloride resulted in the most significant reduction in total phosphorus content. This process releases a large amount of free Ca^{2+} ions, which then bind to other ligands, such as casein phosphoserine clusters in micelles, explaining the decrease in soluble calcium concentration. Similar displacements of phosphate ions from the aqueous phase to the micellar phase have been observed after zinc or iron additions to milk (Shilpashree et al., 2018). Additionally, casein's phosphoserine clusters (αs - $\beta\text{-CN}$) may develop multi-sites of metal coordination, allowing concurrent $\text{Ca}^{2+}/\text{Fe}^{2+}$ binding without displacement, due to the different preference of calcium and iron binding to amino acids (Guo et al., 2014). Meanwhile, direct binding of iron to the micellar phase could induce a solubilization of calcium ions from the micellar phase to the aqueous phase and then, the formation of calcium citrate and calcium-phosphate, which could return to the micellar phase. However, citrate concentration was not monitored in this study but will be considered in future studies.

3.4. Particle size and zeta potential analysis

Z-average particle size analysis revealed remarkable colloidal stability in both control and iron-fortified skim milk systems, with no significant variations observed across iron concentrations up to 15 mmol/L (Fig. 1A). This maintenance of micellar dimensions (average diameter of 250 nm) strongly indicated preserved structural integrity of casein micelles, consistent with previous reports on mineral-fortified dairy systems (Mittal et al., 2015). The observed size invariance suggested

effective compensation between iron-induced charge neutralization and calcium-mediated stabilization mechanisms. Surface charge analysis revealed consistent negative ζ -potential values, ranging from -11 mV to -7 mV across all samples, with significant iron-induced modulation (Fig. 1B). The colloidal stability of casein micelles arose from the equilibrium between electrostatic repulsion and hydrophobic interactions. Divalent cation introduction typically disrupts this balance. Normally, casein micelles in milk are negatively charged. The addition of iron ions leads to electrostatic attraction and charge neutralization. Furthermore, calcium phosphate nanoclusters and the equilibrium between colloidal and serum calcium contribute to micelle stability. Ferrous ions binding to negatively charged phosphoserine residues also reduces the overall charge. Differences in the dissociation ability of the iron salts affect the number of iron ions available for binding, leading to variations in ζ -potential values. Skim milk with the addition of ferrous chloride exhibited the lowest potential values, due to the higher dissociation of iron ions, resulting in more iron ions binding to casein micelles (Table 1). In addition, the changes in mineral distribution and zeta potential may cause changes in thermal stability of samples, future studies should specifically investigate the performance of these iron-fortified systems under industrial heat treatments to fully assess their technological feasibility and commercial potential.

3.5. Flow behavior and viscosity analysis

Immediately after the addition of the iron salt to the skim milk sample, the flow behavior was monitored with increasing shear rate from 0.010 to 100 s^{-1} . The flow curves for skim milk samples with iron salt concentrations between 0 and 15 mmol/L are shown in Fig. 2A-F. A power law (Ostwald-de Waele model) applied to shear stress versus shear rate was used to determine the consistency coefficient (K), and flow behavior index (n) (Anema et al., 2014):

$$\tau = K\dot{\gamma}^n \quad \tau = K\dot{\gamma}^n \quad (1).$$

Where τ is the shear stress (Pa), $\dot{\gamma}$ is the shear rate ($1/\text{s}$), K is the consistency coefficient ($\text{Pa}\cdot\text{s}^n$) and n is the flow behavior index. The results are shown in Table 2. Skim milk with low iron addition behaves as a Newtonian fluid. At concentration levels exceeding 5 mM , a non-linear relationship between shear stress and shear rate was observed consistent with the results of (Zhao et al., 2023), indicating a shear-thinning behavior, likely due to the regular arrangement of macromolecules (Lam et al., 2018). An increase in the final apparent viscosity was observed with higher iron salt concentrations, with little differences among the different iron salts (Table 2). This increase in viscosity may be attributed to: the transition of soluble calcium to the micellar phase, enhancing

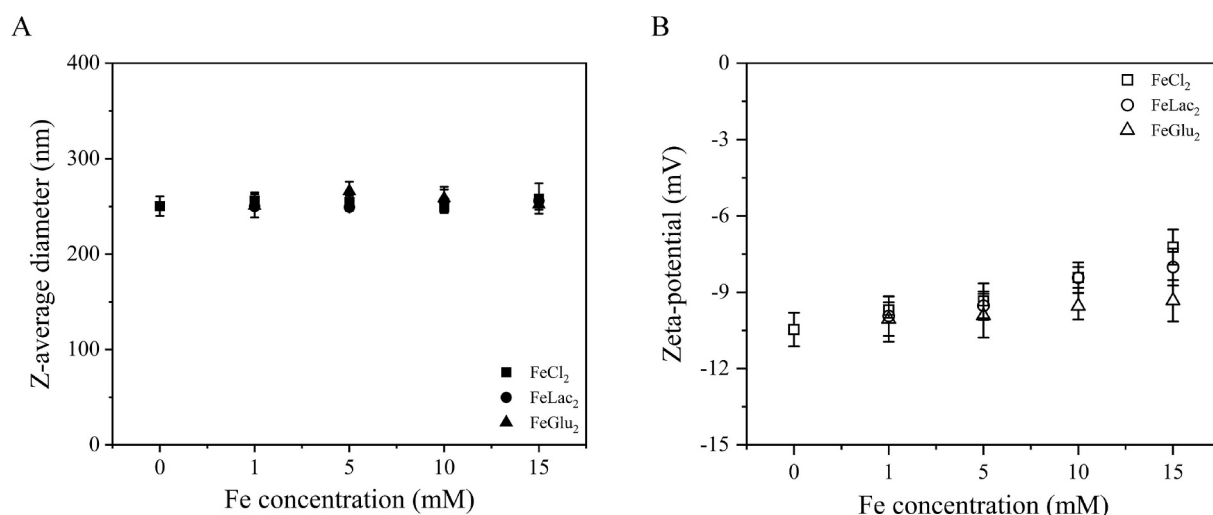


Fig. 1. Effect of iron addition on the hydrodynamic mean (Z-average) diameter and ζ -potential of skim milk. Data points represent means $\pm$ SD from three replicates.

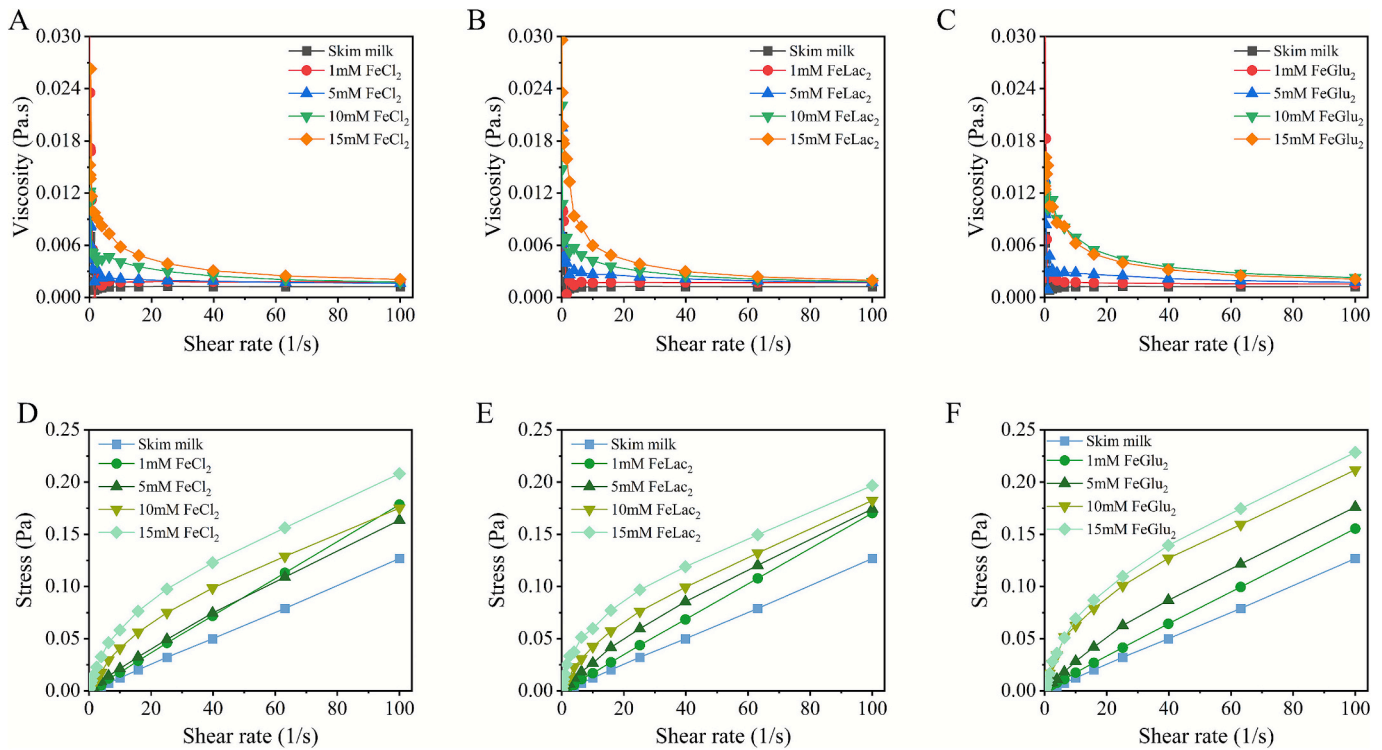


Fig. 2. The viscosity or shear stress versus shear rate at 20 ± 1 °C for 10 % (w/w) total solid skim milk with added with different concentrations of (A/D) ferrous chloride, (B/E) iron(II) lactate and (C/F) iron(II) gluconate.

Table 2

Ostwald-de Waele model parameter values for different concentrations of ferrous chloride, iron(II) lactate and iron(II) gluconate. K, Flow behavior index, n , flow behavior index, and η_{100} , the shear viscosity in 100 1/s.

Sample	K	n	η_{100} (Pa·s)
Skim milk	$(1.23 \pm 0.05) \times 10^{-3}$	1.00614 ± 0.01141	$(1.27 \pm 0.02) \times 10^{-3}$
1 mM FeCl ₂	$(1.5 \pm 0.2) \times 10^{-3}$	1.03726 ± 0.02956	$(1.67 \pm 0.14) \times 10^{-3}$
5 mM FeCl ₂	$(3.0 \pm 0.1) \times 10^{-3}$	0.87049 ± 0.000847	$(1.69 \pm 0.08) \times 10^{-3}$
10 mM FeCl ₂	$(8.6 \pm 0.5) \times 10^{-3}$	0.65686 ± 0.01533	$(1.73 \pm 0.02) \times 10^{-3}$
15 mM FeCl ₂	$(1.49 \pm 0.07) \times 10^{-2}$	0.57204 ± 0.01205	$(2.0 \pm 0.1) \times 10^{-3}$
1 mM FeLac ₂	$(1.6 \pm 0.1) \times 10^{-3}$	1.01514 ± 0.01985	$(1.67 \pm 0.07) \times 10^{-3}$
5 mM FeLac ₂	$(4.5 \pm 0.2) \times 10^{-3}$	0.79626 ± 0.0083	$(1.76 \pm 0.02) \times 10^{-3}$
10 mM FeLac ₂	$(9.0 \pm 0.4) \times 10^{-3}$	0.65254 ± 0.01111	$(1.82 \pm 0.02) \times 10^{-3}$
15 mM FeLac ₂	$(1.81 \pm 0.09) \times 10^{-2}$	0.51536 ± 0.0119	$(1.88 \pm 0.11) \times 10^{-3}$
1 mM FeGlu ₂	$(1.3 \pm 0.2) \times 10^{-3}$	1.0396 ± 0.03121	$(1.66 \pm 0.09) \times 10^{-3}$
5 mM FeGlu ₂	$(4.8 \pm 0.3) \times 10^{-3}$	0.7814 ± 0.0129	$(1.77 \pm 0.02) \times 10^{-3}$
10 mM FeGlu ₂	$(1.65 \pm 0.08) \times 10^{-2}$	0.56391 ± 0.01571	$(1.97 \pm 0.10) \times 10^{-3}$
15 mM FeGlu ₂	$(1.7 \pm 0.1) \times 10^{-2}$	0.55375 ± 0.01224	$(2.04 \pm 0.11) \times 10^{-3}$
Type of salt	NS	NS	NS
concentration	*	*	*
Interaction	NS	NS	NS

NS: no significant effect ($p > 0.05$); *: significant effect ($p < 0.05$).

casein molecule cross-linking and resulting in tighter micelle structures or larger aggregates (Pougher et al., 2024), the charge neutralization effect of iron weakening electrostatic repulsion between casein micelles (Zhao et al., 2023), and the higher charge density of iron, facilitating cross-linked network formation with protein carboxyl or phosphate groups (Jing et al., 2017). This promotes the formation of weak, transient protein interactions, leading to a percolating network structure at rest that impedes flow and explains the observed increase in viscosity and shear-thinning behavior. It is important to note that this network is reversible and is disrupted by dilution and shear, which is why no significant change in the hydrodynamic diameter of individual micelles was detected by dynamic light scattering. Similar increases in milk viscosity with added zinc salts, attributed to specific salt ion interactions with casein micelles, have been reported (Rana et al., 2018).

3.6. Mechanisms of calcium bioavailability in iron-fortified skim milk hydrolysates

3.6.1. Calcium absorption in iron-fortified skim milk hydrolysates

To evaluate calcium bioavailability in iron-fortified skim milk, integrated in vitro gastrointestinal digestion and intestinal absorption models were established. To maintain cellular osmotic balance, the iron-fortified milk hydrolysates (Fe-MH) were diluted 4-fold, yielding a physiologically relevant calcium concentration of approximately 1 mM (Perales et al., 2005, 2006). Comparative analysis revealed that skim milk hydrolysates exhibited 1.8-fold higher calcium absorption efficiency than equimolar CaCl₂, and the results are shown in Table 3. This increased efficiency is attributable to bioactive peptides generated during enzymatic proteolysis. Bovine milk proteins, including whey proteins and caseins, produce bioactive peptides through enzymatic (e. g., pepsin, trypsin) hydrolysis (Korhonen, 2009). Casein-derived phosphopeptides (CPP) and whey-derived peptides demonstrate unique mineral chelation capacities through phosphate-mediated hydration shells, preserving structural stability (Wu et al., 2024). Specifically, CPP is recognized as effective carriers or solubilizers of mineral nutrients.

Table 3

Calcium absorption and cytosolic Ca^{2+} influx rate in Caco-2 cells exposed to different concentration of iron-fortified skim milk hydrolysates (Fe-MH) from different iron salts for 2 h. 1 mM calcium chloride was used as a positive control, equivalent to the calcium concentration in Fe-MH and natural skim milk hydrolysate was used as negative control. Calcium absorption was quantified by measuring the calcium concentration in the basolateral chamber using ICP-OES. Cytosolic Ca^{2+} influx rates are expressed as changes in fluorescence values over time, measured using the calcium-sensitive fluorescent dye Fluo-3 AM.

Samples	Calcium absorption (%)	Cytosolic Ca^{2+} influx rate ($[\text{F}/\text{F}_0]/\text{min}$)
1 mM CaCl_2	3.67 ± 0.70	$(7.7 \pm 0.2) \times 10^{-3}$
Skim milk	6.58 ± 1.60	$(3.3 \pm 0.2) \times 10^{-3}$
1 mM FeCl_2	6.19 ± 0.99	$(3.0 \pm 0.1) \times 10^{-3}$
5 mM FeCl_2	6.18 ± 1.49	$(3.3 \pm 0.2) \times 10^{-3}$
10 mM FeCl_2	5.50 ± 0.95	$(3.25 \pm 0.08) \times 10^{-3}$
15 mM FeCl_2	4.87 ± 1.08	$(3.3 \pm 0.1) \times 10^{-3}$
1 mM FeLac_2	6.28 ± 1.38	$(3.2 \pm 0.1) \times 10^{-3}$
5 mM FeLac_2	5.46 ± 0.99	$(2.8 \pm 0.2) \times 10^{-3}$
10 mM FeLac_2	5.11 ± 0.41	$(3.3 \pm 0.2) \times 10^{-3}$
15 mM FeLac_2	5.01 ± 0.34	$(3.2 \pm 0.3) \times 10^{-3}$
1 mM FeGlu_2	6.52 ± 0.45	$(2.7 \pm 0.1) \times 10^{-3}$
5 mM FeGlu_2	6.31 ± 1.36	$(2.9 \pm 0.1) \times 10^{-3}$
10 mM FeGlu_2	6.25 ± 1.03	$(3.3 \pm 0.3) \times 10^{-3}$
15 mM FeGlu_2	6.19 ± 0.51	$(3.3 \pm 0.1) \times 10^{-3}$
Type of salt	*	NS
Salt concentration	NS	*
Interaction	NS	NS

NS: no significant effect ($p > 0.05$); *: significant effect ($p < 0.05$).

CPP has been reported to enhance calcium bioavailability by forming soluble complexes with calcium ions via phosphoserine coordination, thereby improving intestinal lumen retention and epithelial transport (Miquel et al., 2006; Wang et al., 2023). The soluble iron contents in Fe-MH are around 15–110 μM (Table S1) which is influenced by the adding concentration and type of iron salts as well as their interactions. Soluble iron concentration increased with higher additions of all three iron salts in Fe-MH ($p < 0.05$). For three types of iron salts, milk fortified with ferrous chloride exhibited lower soluble iron concentrations after digestion compared to that fortified with iron(II) gluconate at higher fortification concentrations ($> 5 \text{ mM}$). However, the differences between soluble iron concentration did not affect calcium absorption rates significantly. Linear regression analysis showed that skim milk fortified with iron(II) gluconate exhibited the highest calcium absorption, followed by iron(II) lactate and ferrous chloride, correlating with anion-specific chelation dynamics. Iron(II) gluconate, in particular, showed minimal inhibition of calcium absorption, even at high concentrations. Gluconate's dual affinity for $\text{Ca}^{2+}/\text{Fe}^{2+}$ facilitates the formation of CaGlu^+ complexes that are favorable for absorption, whereas chloride's

weak coordination allows free Fe^{2+} to interfere with calcium transporters.

3.6.2. Kinetics of intracellular calcium uptake from iron-fortified skim milk hydrolysates

To elucidate the kinetic behavior of calcium uptake from iron-fortified skim milk hydrolysates, a continuous increase in Fluo-3 fluorescence intensity, a fluorescent calcium indicator, was observed over time (Cegarra et al., 2022). This continuous increase in fluorescence intensity indicates the dissociation process of calcium ions, and the results are shown in Fig. 3. Real-time calcium flux monitoring via Fluo-3 fluorescence revealed kinetic divergences between calcium absorption and cytosolic Ca^{2+} influx rates (Table 3). While CaCl_2 displayed a rapid fluorescence increase with an uptake rate of $(7.7 \pm 0.2) \times 10^{-3} [\text{F}/\text{F}_0]/\text{min}$, reflecting free ion diffusion, Fe-MH samples showed gradual signal accumulation, potentially indicating complex competitive binding reactions (Cui et al., 2015). Because most calcium in milk is bound to peptides or amino acids, these complexes can be transported via calcium transporters, peptide transporters, or vesicular transport. The binding status may be unfavorable for Fluo-3 binding. As shown in Fig. 3, significantly lower cytosolic Ca^{2+} influx rates were observed for skim milk after 30 min of exposure compared with calcium chloride, further corroborating preferential cellular internalization of peptide-complexed calcium over free ions, suggesting calcium may be bound to peptides rather than Fluo-3 (Phummisutthigoon et al., 2022). The calculated cytosolic Ca^{2+} influx rates for the different Fe-MH samples are presented in Table 3. The results indicate no significant difference in the rate of calcium uptake among the Fe-MH samples, but a slight increase at high iron concentrations, attributed to iron binding to peptides and releasing calcium, consistent with the calcium distribution results.

3.7. Buffering effects of milk protein hydrolysates on calcium absorption in the presence of iron salts

3.7.1. Cell viability

Cell viability experiments showed that milk hydrolysates (MH) had no significant effect on cell viability at low concentrations (0.01–0.3 mg/mL), as shown in Fig. 4A. As the concentration of milk hydrolysates increased (0.4–1 mg/mL), MH promoted an increase in cell viability, which may contribute to its role in attenuating the negative effect of iron on calcium absorption (Sun et al., 2023). However, when the concentration exceeded 1 mg/mL, cell viability decreased, suggesting a threshold effect and cytotoxicity.

3.7.2. Cellular antioxidant activity of iron-fortified skim milk hydrolysates

The observed stability in calcium absorption within iron-fortified skim milk systems likely stems from the antioxidant properties

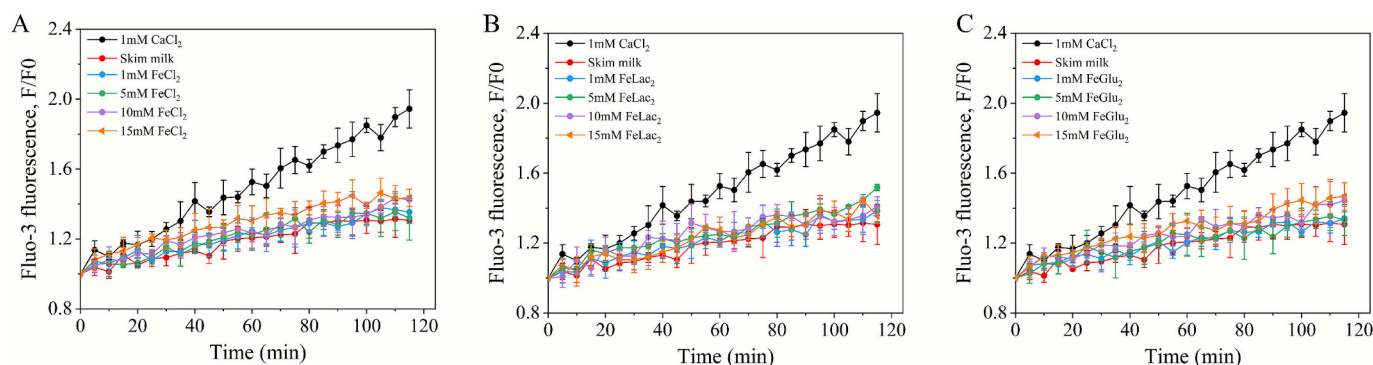


Fig. 3. Temporal changes in intracellular calcium concentration in Caco-2 cells exposed to iron-fortified skim milk hydrolysates (Fe-MH). The skim milk was fortified with (A) ferrous chloride, (B) iron(II) lactate and (C) iron(II) gluconate. Calcium chloride containing the same calcium concentration compared with Fe-MH was used as a positive control. The y-axis represents the relative fluorescence intensity (F/F_0), where F is the fluorescence value at a given time point and F_0 is the mean of the initial fluorescence measurements. Data points represent means $\pm$ SD from four replicate wells.

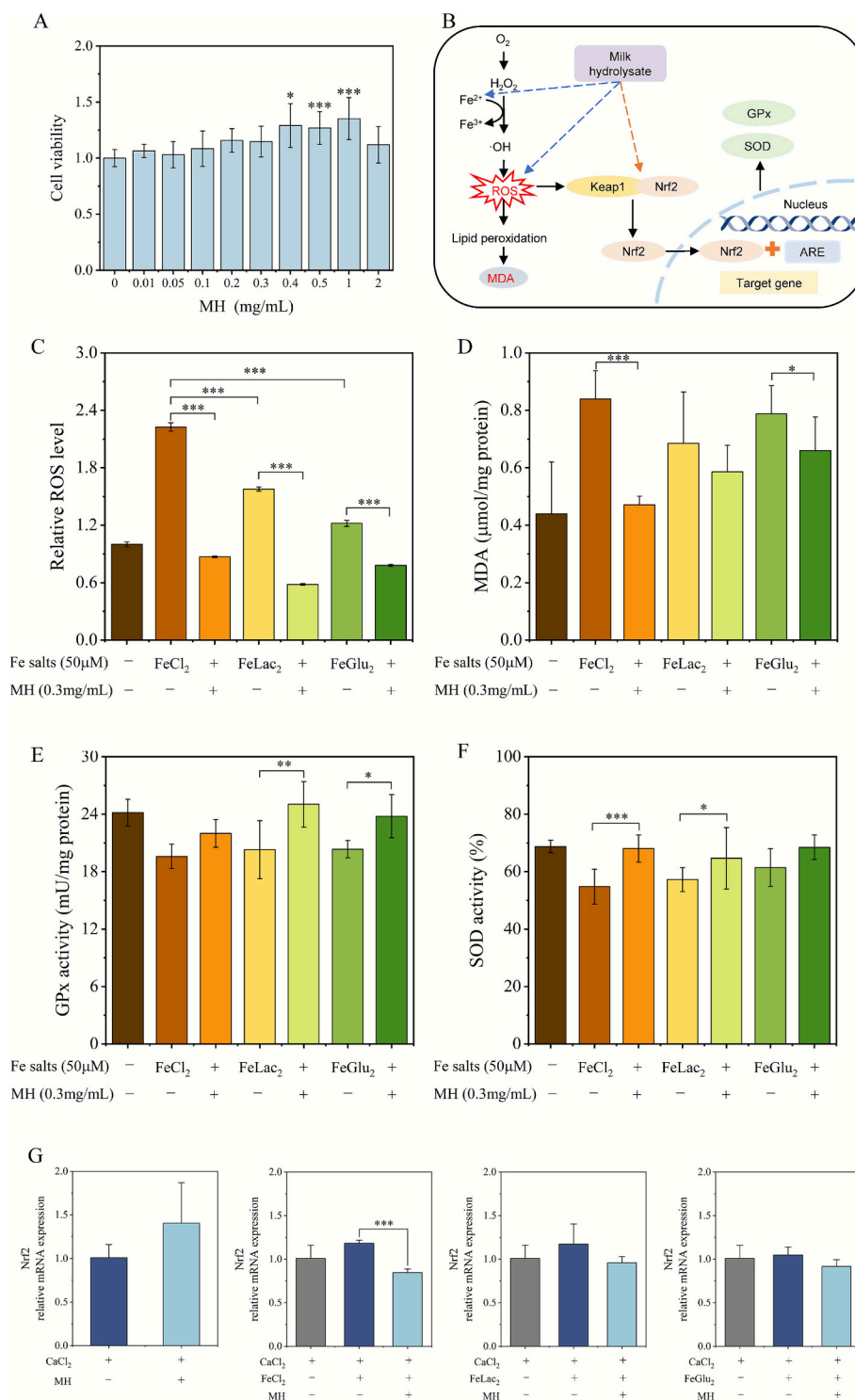


Fig. 4. Mechanism of mitigation of the inhibitory effect of iron salts on calcium absorption by milk hydrolysates (MH). (A) Effect of MH addition on the viability of Caco-2 cells. (B) Intervention of MH on the course of iron-induced oxidative stress responses. MH binding to Fe^{2+} reduces ROS generated by the Fenton reaction, stabilizes the binding of Keap1 with Nrf2, and slows iron overload-induced Nrf2 activation. Effect of MH on iron salt-induced (C) relative ROS levels and (D) lipid peroxidation (MDA) at 24 h with DMEM as control. Effect of adding iron salts with and without MH on GSH (E) and SOD (F) enzyme activities in Caco-2 cells. (G) Relative mRNA expression levels of Nrf2 in Caco-2 cells treated with ferrous chloride and MH for 24 h. Data are presented as mean $\pm$ SD from six independent experiments ($n = 6$). Asterisks indicate significant differences compared to the control: * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$.

inherent to milk hydrolysates (MH). While iron supplementation is known to impair cellular calcium uptake through oxidative mechanisms (Li et al., 2025), our investigation employing 0.3 mg/mL MH revealed a multifaceted protective capacity against redox imbalance, as schematically illustrated in Fig. 4B. Cellular metabolism naturally generates

reactive oxygen species (ROS) - including superoxide (O_2^-), hydrogen peroxide (H_2O_2), and hydroxyl radicals ($OH^\cdot$) - which at physiological levels mediate critical signaling functions and immune defense but become cytotoxic when overproduced (Park & Chung, 2019).

As shown in Fig. 4C, exposure to 50 μ mol/L iron salts individually for

24 h modulated ROS generation and the elevated level of ROS disrupts the cellular redox homeostasis, leading to oxidative stress in vivo, and ultimately causing oxidative damage to the cells (Diaz De Barboza et al., 2017; Nakamura et al., 2019). Ferrous chloride provoked the most dramatic oxidative increase, elevating intracellular ROS 2.2-fold above baseline, whereas gluconate and lactate formulations exhibited weaker effects (Kraidith et al., 2016). MH intervention effectively restored redox equilibrium through dual pathways: direct binding with free Fe^{2+} , scavenging ROS and suppressing iron chelation-mediated Fenton reaction (Power et al., 2013). An increased generation of ROS during oxidative stress initiates a chain reaction of degradation steps termed lipid peroxidation. Malondialdehyde (MDA) is a by-product of lipid peroxidation, which shows cytotoxicity, mutagenicity and carcinogenicity, and serves as a biomarker indicating the degree of oxidative damage to cells (Busch & Binder, 2017). Fig. 4D showed that the iron salts resulted in elevated levels of MDA, and addition of MH protected against iron salt-induced lipid peroxidation by up to 50 %, paralleling its protection against hydrogen peroxide toxicity (Wang et al., 2025).

The cellular antioxidant defense system responded synergistically to MH exposure. As shown in Figs. 4E-F, MH treatment significantly enhanced superoxide dismutase (SOD) and glutathione peroxidase (GSH-Px) activities ($p < 0.05$). SOD catalyzes superoxide radical dismutation, while GSH-Px maintains redox balance through hydrogen peroxide reduction and glutathione regeneration (Park et al., 2021; Sanjukta et al., 2021). These findings align with established mechanisms of milk protein modulating iron redox activity (Caetano-Silva et al., 2021). Notably, the hierarchy of iron salt effects on oxidative parameters ($\text{FeCl}_2 > \text{FeLac}_2 > \text{FeGlu}_2$) inversely correlated with calcium absorption efficiencies, suggesting oxidative stress mitigation as a pivotal regulator of mineral uptake dynamics. The differential behavior originates from anion-specific iron coordination capacities - strongly chelating gluconate limits free Fe^{2+} availability, thereby reducing both oxidative damage and calcium absorption interference, whereas weakly-bound chloride permits maximal iron reactivity.

Nuclear factor erythroid 2-related factor 2 (Nrf2) is a key transcription factor that regulates cellular antioxidant responses. It regulates the expression of various antioxidant genes through the activation of antioxidant response elements (AREs), which help cells to cope with oxidative stress and toxic substances. Kelch-like ECH-associated protein 1 (Keap1), a redox-sensitive signaling rheostat, maintains Nrf2 in an inactive state by sequestering it in the cytoplasm through constitutive binding under physiological conditions (Ulasov et al., 2022). Our findings demonstrate iron overload significantly activates Nrf2 mRNA expression (Fig. 4G), consistent with the mechanisms where Fe^{2+} -generated ROS modify reactive cysteine residues (particularly Cys151) in Keap1's sensor domain (Lim et al., 2019). This post-translational modification disrupts Keap1-Nrf2 binding, enabling Nrf2 nuclear translocation and subsequent upregulation of antioxidant enzymes and the mechanisms shown in Fig. 4B. Notably, MH treatment elevated Nrf2 expression by 38 % versus controls without iron, corroborating previous reports on bioactive milk peptides' antioxidant potential (Tonolo et al., 2018, Tonolo et al., 2020). However, co-treatment with iron salts and MH stabilized Nrf2 mRNA expression close to baseline level may through dual modulation: ROS scavenging reduces Nrf2 activation signals and MH modifies the thiol group modification of Keap1 stabilizing the cytoplasmic Keap1-Nrf2 complex (Wang et al., 2022). This paradoxical regulation suggests MH's bifunctional antioxidant capacity - potentiating basal Nrf2-mediated defenses while preventing excessive activation under iron challenge. The integrated antioxidant effects of MH include: reduction in ROS generation, suppression of lipid peroxidation (MDA), enhancement of SOD/GSH-Px activities and stabilization of Nrf2 expression. These protective mechanisms, schematized in Fig. 4B, similar to findings from bass fish hydrolysates and eggshell membrane hydrolysates, indicating a regulatory mechanism that maintains cellular redox poise without triggering compensatory over-expression of antioxidant systems (Li et al., 2024; Shi et al., 2014).

3.8. Mechanisms of calcium absorption in iron-fortified skim milk

To further understand the mechanisms underlying these observations, we conducted additional experiments examining the mRNA expression and activity of key calcium transporters in the presence of iron salts and MH, as shown in Fig. 5. Calcium absorption operates through two synergistic pathways: the classical transcellular/paracellular system mediated by TRPV6 and Claudin-2 transporters, and the peptide-assisted transport of MH-calcium complexes (Li et al., 2025). The results showed that the presence of iron salts inhibited the mRNA expression of the calcium transporters, especially in *Calbindin-D9k* and *PMCA1b* while MH intervention restored calcium absorption efficiency through transcriptional regulation. The addition of MH significantly increased the mRNA expression of *TRPV6* and *Claudin-2* and restored the decreased expression caused by iron salts (Fig. 5A-B), may be attributed to the fact that calcium can be released from the peptide-calcium chelates when it reaches the small intestine prior to uptake (Guo et al., 2014). This mechanism aligns with prior observations of CPP-Ca complex dissociation increasing the concentration of soluble calcium and enhancing passive absorption in the distal ileum (Lee et al., 1980).

Peptide-assisted transport of MH-calcium complexes involves a sophisticated transport network encompassing intercellular TJs (tight junctions), PepT1 (Proton Coupled Oligopeptide Transporter 1)-mediated osmosis, vesicle-mediated intracellular transport of oligopeptides (transcytosis), and transcellular endocytosis across intestinal epithelium involving transport activity based on the peptide's hydrophobicity, structure and molecular size (Zhao et al., 2025). Calcium bound to MH may be transported by these pathways in Caco-2 cells. PepT1 is a high-affinity proton-dependent oligopeptide transporter protein that transports digested oligopeptides from the intestinal lumen to intestinal epithelial cells. Our findings demonstrate MH-bound calcium utilizes PepT1-mediated pathways (1.4-fold mRNA upregulation) and vesicular transport systems, evidenced by stable *PMCA1b* expression suggesting alternative efflux routes (Fig. 5C-E). However, we found a negative effect of MH on *Calbindin-D9k* mRNA expression, probably because MH-peptide complexes buffer intracellular free calcium, attenuating calcium-sensing receptor signaling. In addition, larger peptides may be absorbed intact through intestinal barriers such as vesicles, as in the case of the 17-residue peptide from β -CN, where the peptide is combined with metal ions, and the metal ions can then be absorbed by the same pathway (Regazzo et al., 2010). García-Nebot et al. also suggested that zinc bound to the CPP can be transported through the membrane of Caco-2 cells, and this phenomenon is attributed the positive effect of β -CN(1–25)4P on zinc uptake (García-Nebot et al., 2013).

Furthermore, the addition of MH can also regulate iron absorption. Parallel iron absorption studies revealed MH-iron chelates exhibit superior solubility (90 %) compared with FeSO_4 (5.9 %) at pH 6–8. Fe-MH transport utilizes peptide transport pathways including vesicular endocytosis. Moriya and Linder found that using specific inhibitors of vesicular trafficking, including nocodazole and 5'-(4-fluorosulfonylbenzoyl)-adenosine could reduce iron uptake and basolateral transport up to 50 % (Moriya & Linder, 2006). Moreover, β -CPP bound iron was also shown to be absorbed through endocytosis (Péres et al., 1999). Iron overload (50 $\mu\text{mol/L}$, 24 h) downregulated *DMT1*/*FPN1* expression (Fig. 5F-G), with ferrous chloride showing strongest inhibition (60 %) due to higher free Fe^{2+} bioavailability suggesting a compensatory or negative feedback response during excessive iron uptake (Tallkvist et al., 2000). The results were similar to previous studies in that cell exposure to iron (200 $\mu\text{mol/L}$ ferric nitrilotriacetate for 72 h) significantly reduced *DMT1* mRNA (57 %) and *FPN1* mRNA (52 %) expression (Martini et al., 2002). However, *FPN1* mRNA expression was up-regulated with the addition of MH. The changes may be due to the antioxidant effect of MH. The *FPN1* promoter region contains the antioxidant response element (ARE) which can be activated by Nrf2. It was shown that *FPN* expression was significantly reduced in Nrf2 knockout mice, leading to iron accumulation in tissues (e.g., liver, small intestine)

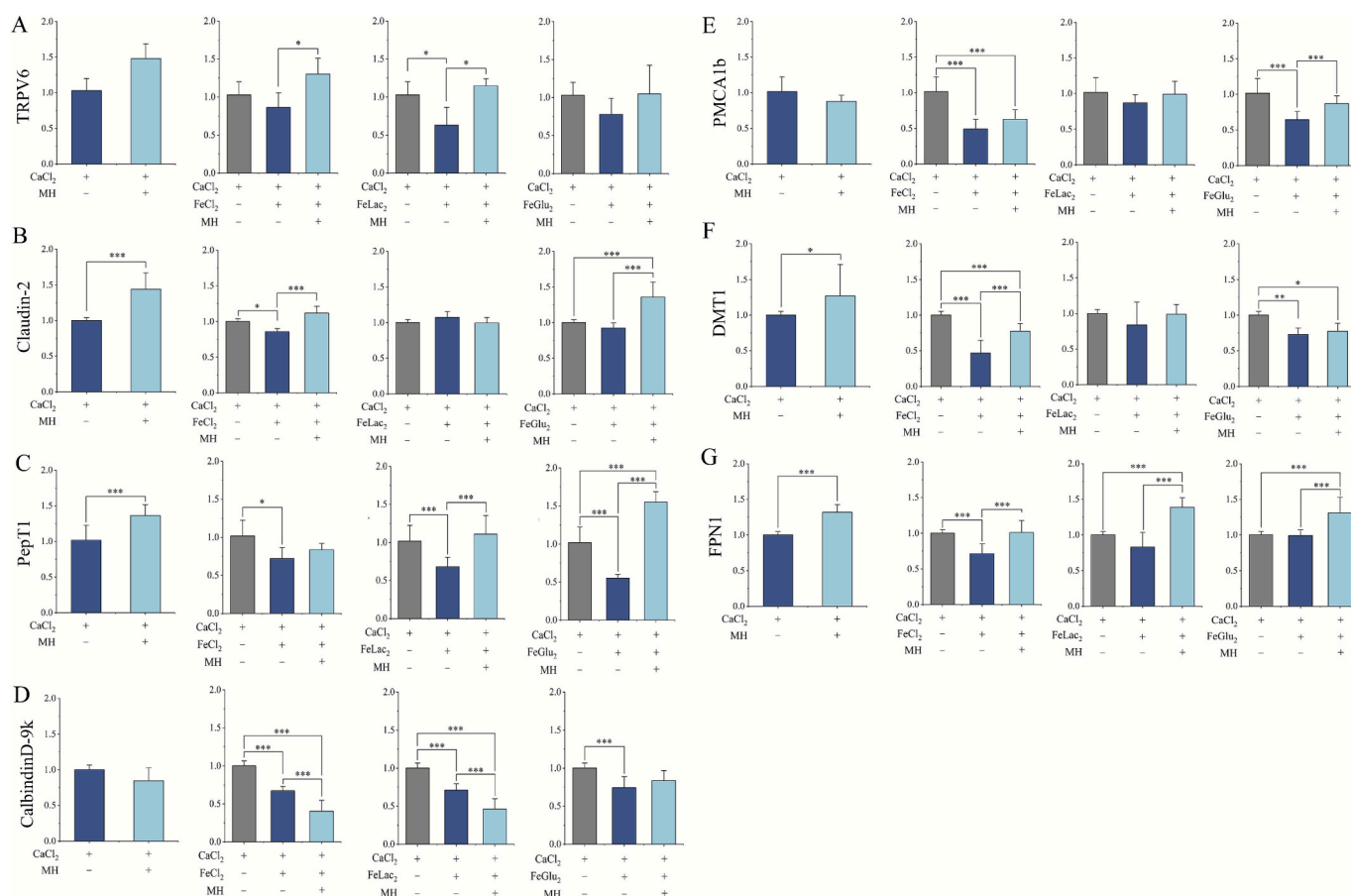


Fig. 5. Gene expression analysis of calcium and iron transport-related genes in Caco-2 cells treated with calcium chloride, ferrous salts and milk hydrolysates (MH). Relative mRNA expression levels of calcium transport genes: (A) TRPV6, (B) Claudin-2, (C) PepT1, (D) Calbindin-D9k, (E) PMCA1b, and iron transport genes: (F) DMT1, (G) FPN1 in Caco-2 cells. Cells were treated with 1 mM calcium chloride in the presence or absence of 50 μ M ferrous salts and MH for 24 h. Gene expression was quantified using qRT-PCR with β -actin as the reference gene. Data are presented as mean $\pm$ SD from six independent experiments (n = 6). Asterisks indicate significant differences compared to the calcium chloride only control: *P < 0.05, **P < 0.01, ***P < 0.001.

(Liu et al., 2020). In hepatocyte (LO2) cells, the antioxidant ZSF (Qizhufang, a Chinese herbal formula) significantly reversed the inhibitory effect of iron overload on FPN1 through activation of Nrf2, while up-regulating Ferritin and Copper Cyanin (CP), creating a synergistic protection of iron storage and export (Xie et al., 2019). Also, *DMT1* mRNA expression increased slightly, consistent with the results of Du (Fen, 2016) while Argyri showed that although milk peptides enhanced iron uptake in Caco-2 cells, they had no effect on *DMT1* expression (Argyri et al., 2009). The regulatory mechanism of MH on key genes expression is shown in Fig. S2.

4. Conclusion

In conclusion, our study reveals that iron fortification produces a dose-dependent effect on the physicochemical properties of skim milk, with a decrease in soluble calcium and phosphorus content but increase in calcium activity. The partitioning of ions between the two phases is mainly related to the complexing ability of the anionic ligands. In vitro simulations of gastrointestinal digestion showed that iron-fortified milk did not interfere with calcium absorption but decrease free calcium in Caco-2 cytoplasm. Integrated transport model reveals the effects of Fe-MH on calcium absorption mechanisms of: modulating the proliferation and differentiation of intestinal epithelial cells, activating peptide-calcium complex transporters, mitigating the oxidative stress induced by iron salts, and regulating calcium transporter expression regulation. Simultaneously, MH enhances iron bioavailability through solubility preservation, peptides' transport pathway utilization, and

Nrf2/FPN1 regulatory network activation. Thus, skim milk is a multi-functional and high-quality carrier for iron fortification and facilitates synchronized supplementation of calcium and iron.

CRediT authorship contribution statement

Yi Li: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yongqiang Cheng:** Supervision, Resources, Funding acquisition. **Ning Tang:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they do not have any known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.foodchem.2025.147462>.

Data availability

Data will be made available on request.

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